

# Module 6 – RAG Quick Revision

## #RAG Basics

- Retrieves relevant documents before LLM generation.
- Reduces hallucinations using external knowledge.
- Keeps responses grounded in latest data.
- Ideal for enterprise QA and chatbots.

## #Document Loading

- Load PDF, DOCX, TXT, HTML, CSV, Excel.
- Normalize extracted text.
- Preserve metadata.
- Clean noisy content.

## #Chunking

- Split documents into searchable pieces.
- Recursive chunking is the best general-purpose strategy.
- Choose proper chunk size and overlap.
- Preserve semantic meaning.

## #Embeddings

- Convert text into vectors.
- Similar meaning → similar vectors.
- Used for semantic search.
- Generated once and stored.

## #Vector Store

- Store embeddings with metadata.
- Chroma is simple for local projects.
- Supports similarity search.
- Enables fast retrieval.

## #Retrieval

- Embed user query.
- Retrieve Top-K similar chunks.
- Apply threshold and metadata filters.
- Return only relevant context.

## #Hybrid Search

- Combine BM25 + Vector Search.

- Improves keyword and semantic matching.
- Use RRF for fusion.
- Better recall than either alone.

## #Reranking

- Reorders retrieved chunks.
- Cross-encoders improve accuracy.
- Removes irrelevant context.
- Send only best chunks to LLM.

## #Prompt Building

- Inject retrieved context.
- Instruct model to answer only from context.
- Handle missing information gracefully.
- Include citations if required.

## #Evaluation

- Measure Faithfulness.
- Measure Context Precision/Recall.
- Measure Answer Relevancy.
- Use RAGAS/TruLens.

## Key Parameters

- Chunk Size: 256-1024 (512/1000 common)
- Chunk Overlap: 10-20%
- Top-K: 3-10
- Similarity Threshold: 0.25-0.45
- Embedding Models: all-MiniLM-L6-v2, BGE-small/base
- Reranker: FlashRank, BGE Cross Encoder
- Hybrid Search: BM25 + Vector Search + RRF
- Metrics: Faithfulness, Context Recall, Context Precision, Answer Relevancy, MRR, Hit@K